

Gene Target Validation Dossier

Target gene: TRPC6 **Disease:** Focal Segmental Glomerulosclerosis **Generated:** 2026-06-24 03:57 UTC

Evidence: 135 kept / 67 dropped

Full evidence report: [full_report.md](#) — categorized kept sources with external links

Executive Summary

✓ **Overall consensus: Support** (confidence 86%) | **Suitability score: 85/100**

Lens Summary

- **Genetics lens** ✓: Support (90%) — The genetic evidence strongly supports TRPC6 as a causal driver of Focal Segmental Glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, and Mendelian-grade validation via ClinVar. The constraint profile indicates that the gene is LoF-tolerant but missense intolerant, supporting an inhibitory therapeutic approach. [△ CONSTRAINT GUARD: The text claims missense criticality/constraint/intolerance, but $\text{mis_z}=1.70$ (no meaningful missense constraint), $\text{missense OEUF}=0.928$ (missense-tolerant (gene accepts amino-acid change globally)) — the gene does not show significant global missense constraint. Pathogenic missense variants may still exist at specific residues (GoF), but the gene is not globally missense-intolerant.]
- **Biology lens** ✓: Support (90%) — The evidence strongly supports TRPC6 as a mechanistically understood and druggable target for FSGS inhibition, with clear functional studies linking it to podocyte injury and glomerular disease.
- **Clinical lens** ✓: Support (80%) — Clinical trials and preclinical studies provide strong evidence that TRPC6 inhibition is a viable therapeutic strategy for FSGS. The phase II clinical trial of BI 764198, a TRPC6 inhibitor, in people with FSGS (NCT05213624) has completed, suggesting the target's relevance to the disease. Preclinical studies show that TRPC6 knockout or pharmacological inhibition can reduce kidney disease severity and proteinuria in various models.
- **Safety lens** ✓: Support (85%) — The target TRPC6 shows no curated Open Targets safety liabilities and is associated with Focal Segmental Glomerulosclerosis (FSGS) through multiple genetic associations, including pathogenic variants in ClinVar. Mouse KO phenotypes include cardiovascular effects such as increased vasoconstriction and abnormal vascular smooth muscle physiology, which are candidate on-target liabilities for TRPC6 inhibition. The gene is relatively LoF-tolerant with no homozygous carriers observed in gnomAD, supporting the safety of pharmacological inhibition. Expression levels are low but uninformative in bulk kidney tissue due to podocyte-specificity, and the target is expressed at high levels in other tissues like lung and esophagus. The therapeutic direction aligns with the known gain-of-function mechanism, where TRPC6 mutations cause increased calcium influx and podocyte injury.
- **Commercial lens** ✓: Support (85%) — The IP landscape is moderately encumbered but not prohibitive, with 17 patents focused on TRPC6. The competitive field is underserved by existing drugs, as no approved or clinical-stage therapies targeting this gene were found. The market size for hereditary steroid-resistant nephrotic syndrome (FSGS) is unknown from Orphanet, but the indication is rare and likely suitable for orphan drug designation. [△ COMMERCIAL GUARD:

'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

- **Regulatory lens ?**: Insufficient evidence (0%) — No evidence of this lens's type passed screening; verdict reflects the evidence gap, not a negative finding.

Evidence note: Proceed as the completeness scores are generally above 60, and there are no critical gaps in genetics or biology.

Evidence Summary

135 kept / 67 dropped (total: 202)

Kept Evidence

Literature (97)

Source	Detail	Quality	Year	First Author
PMID:42041578	An Inducible hiPSC-Derived Human Podocyte Model for Functional Analysis of TRPC6 Variants Associated with FSGS.	★★★	2026	Batool L
PMID:41616795	TRPC6 inhibition for the treatment of focal segmental glomerulosclerosis: a randomised, placebo-controlled, phase 2 trial of BI 764198.	★★★	2026	Trachtman H
PMID:41481124	Podocyte-specific Translational Profiling In Vivo Uncovers Distinct Patterns in Trpc6-Deficient Podocytes.	★★★	2026	Einloft J
PMID:41100809	Structural and functional insights of the podocyte slit diaphragm complex.	★★★	2026	Qadri AH
PMID:40719843	Targeting TRPC6 in podocytopathies: Why clinical translation remains a challenge?	★★★	2026	Patel CA
PMID:40455255	Phase 1 trials of BI 764198, a transient receptor potential channel 6 inhibitor, in healthy volunteers and participants with kidney impairment.	★★★	2025	Schultz A
PMID:40402558	A randomized, Phase I study of the safety, tolerability, and pharmacokinetics of BI 764198, a transient receptor potential channel 6 (TRPC6) inhibitor, in healthy Japanese men.	★★★	2025	Yonemura T
PMID:40388293	Genotype-phenotype correlations and clinical outcomes of genetic TRPC6 podocytopathies.	★★★	2025	McAnallen SM
PMID:39352759	Natural History and Clinicopathological Associations of TRPC6-Associated Podocytopathy.	★★★	2025	Wooden B

Source	Detail	Quality	Year	First Author
PMID:37615749	An inactivating human TRPC6 channel mutation without focal segmental glomerulosclerosis.	★★★	2023	Batool L
PMID:36007181	Ion channels and channelopathies in glomeruli.	★★★	2023	Staruschenko A
PMID:35920919	Causal and putative pathogenic mutations identified in 39% of children with primary steroid-resistant nephrotic syndrome in South Africa.	★★★	2022	Nandlal L

... and 85 more — see [full_report.md](#)

Empirical (38)

OpenTargets (1)

Source	Detail	Quality
OpenTargets · TRPC6	Open Targets association ENSG00000137672 / EFO_0004236: overall=0.812, genetic=0.956, literature=0.686, affected_pathway=0.000, animal_model=0.000, clinical=0.000, somatic_mutations=0.000.	★★★

Genetics (7)

Source	Detail	Quality
ontology:TRPC6	Ontology constraints for TRPC6: Mode of inheritance: Autosomal dominant (source: HPO/Monarch); HPO phenotype breadth: 7 phenotype(s) (moderate); top terms: Hypertension, Nephrotic syndrome, Proteinuria, Stage 5 chronic kidney disease, Autosomal dominant inheritance.	★★★
gwas_catalog:locus_breadth_summary	TRPC6 locus: 39 distinct GWAS traits found (0 matched Focal Segmental Glomerulosclerosis). Excluded as off-indication: Brain rsFMRI measurement (amplitude of low frequency fluctuations) (Net100_Node28), Brain rsFMRI measurement (amplitude of low frequency fluctuations) (Net100_Node35), Brain rsFMRI measurement (amplitude of low frequency fluctuations) (Net25_Node21), Brain rsFMRI measurement (amplitude of low frequency fluctuations) (Net25_Node16), Brain rsFMRI measurement (amplitude of low frequency fluctuations) (Net100_Node38).	★★★

Source	Detail	Quality
ot_genetics_coloc:locus_breadth_summary	TRPC6 coloc: 6 distinct GWAS traits found (0 matched Focal Segmental Glomerulosclerosis). Excluded as off-indication: Height, Age when periods started menarche, Sleep duration (UKB data field 1160), Smoking initiation, Educational attainment.	★★★
gencc:TRPC6	GenCC gene-disease validity for TRPC6: focal segmental glomerulosclerosis 2 (Strong, Ambry Genetics); focal segmental glomerulosclerosis 2 (Strong, Labcorp Genetics (formerly Invitae)); focal segmental glomerulosclerosis 2 (Strong, PanelApp Australia); familial idiopathic steroid-resistant nephrotic syndrome (Supportive, Orphanet).	★★★
orphanet:TRPC6 (2)	Orphanet associations for TRPC6: Hereditary steroid-resistant nephrotic syndrome (ORPHA:656; Disease-causing germline mutation(s) (gain of function) in; Assessed); Orphanet prevalence: Hereditary steroid-resistant nephrotic syndrome (ORPHA:656): Unknown [Point prevalence, Worldwide, Not yet validated].	★★★
spoke:DOID:1312	SPOKE graph: TRPC6–focal segmental glomerulosclerosis association via DISEASES; gwas_p=n/a, diseases_score=6.393.	★★★

Expression & Omics (19)

Source	Detail	Quality
GTEx/HPA	TRPC6 RNA tissue specificity: Tissue enhanced (Human Protein Atlas).	★★★
UniProt	TRPC6 subcellular localization: Cell membrane (UniProt/HPA).	★★★
GTEx/HPA (14)	Lung: 23.2 TPM, Esophagus_Muscularis: 19.0 TPM, Thyroid: 12.0 TPM, Esophagus_Gastroesophageal_Junction: 9.5 TPM, Adipose_Subcutaneous: 7.9 TPM, Kidney_Cortex: 1.2 TPM, Brain_Cortex: 0.7 TPM, Brain_Cerebellum: 0.6 TPM, Pancreas: 0.3 TPM, Heart_Atrial_Appendage: 0.3 TPM, Heart_Left_Ventricle: 0.2 TPM, Whole_Blood: 0.1 TPM, Liver: 0.0 TPM, GTEx top tissues (median TPM):	★★★

Source	Detail	Quality
	Lung=23.2, Esophagus_Muscularis=19.0, Thyroid=12.0, Esophagus_Gastroesophageal_Junction=9.5, Adipose_Subcutaneous=7.9. HPA specificity: Tissue enhanced. Subcellular: Cell membrane.	
spoke_anatomy:TRPC6	SPOKE knowledge graph confirms TRPC6 expression in 23 anatomical structure(s) (UBERON): Brodmann (1909) area 9, adipose tissue, adrenal gland, adult mammalian kidney, brain, calcaneal tendon, colon, corpus callosum, dorsolateral prefrontal cortex, frontal cortex (+13 more).	★★★
expression_atlas:ensg00000137672	Expression Atlas: no differential expression data specific to 'Focal Segmental Glomerulosclerosis' found for TRPC6; top significant differential results in other contexts: DOWN -5.9-fold (p=0) in 'definitive endoderm cell; 72 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194]; DOWN -4.5-fold (p=0) in 'embryonic stem cell differentiating toward definitive endoderm; 48 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194]; DOWN -4.6-fold (p=6.4e-306) in 'embryonic stem cell differentiating toward definitive endoderm; 36 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194]; DOWN -5.4-fold (p=8.9e-305) in 'embryonic stem cell differentiating toward definitive endoderm; 60 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194]; DOWN -3.0-fold (p=1.4e-234) in 'embryonic stem cell differentiating toward definitive endoderm; 24 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194].	★★★
expression_atlas:E-MTAB-9194	TRPC6 DOWN -3.0-fold (p=1.4e-234) in 'embryonic stem cell differentiating toward definitive endoderm; 24 hour' vs 'embryonic stem cell; 0 hour' [E-MTAB-9194] (Expression Atlas).	★★★

Constraint (3)

Source	Detail	Quality
gnomAD (3)	ClinVar variants in TRPC6 (via gnomAD): 38 Pathogenic (27 missense_variant, 5 synonymous_variant, 3 stop_gained, 2 frameshift_variant, 1 splice_acceptor_variant), 26 Likely Pathogenic (11 missense_variant, 10 frameshift_variant, 4 stop_gained, 1 splice_acceptor_variant), 120 Benign/Likely Benign out of 482 total.; Constraint interpretation for TRPC6: LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)); mis_z=1.70 (no meaningful missense constraint; note: higher mis_z = more missense-constrained); pLI=0.0000 (pLI-tolerant (consistent with LoF tolerance)); missense OEUF=0.928 (missense-	★★★

Source	Detail	Quality
	tolerant (gene accepts amino-acid change globally)). syn_z=-0.000 (≈0 expected; syn_z >2 flags data quality).; gnomAD v4: 194 HC pLoF variants observed in TRPC6. Most common AF=2.63e-05. No homozygous carriers in gnomAD. Absence at this allele count is uninformative — not evidence of biallelic lethality; selection signal comes from LOEUF/o-e.	

Functional Genomics (2)

Source	Detail	Quality
DepMap	DepMap: TRPC6 dependent in unknown lines (Chronos_Combined). Mean Chronos: -0.013 (SD 0.129, Q1/Q3 -0.090/0.067).	★★★
project_score:TRPC6	Project Score: TRPC6 a fitness gene in 348/350 Sanger cell lines (BAGEL2 scaled BF > 0). Mean scaled BF: 3.570.	★★★

Patents (4)

Source	Detail	Quality
16456513	TRPC6 INVOLVED IN GLOMERULONEPHRITIS	★★★
15170916	TRPC6 INVOLVED IN GLOMERULONEPHRITIS	★★★
12782075	TRPC6 INVOLVED IN GLOMERULONEPHRITIS	★★★
11716050	TRPC6 INVOLVED IN GLOMERULONEPHRITIS	★★★

Clinical Trials (1)

Source	Detail	Quality
NCT05213624	A Study to Test BI 764198 in People With a Type of Kidney Disease Called Focal Segmental Glomerulosclerosis	★★★

Regulatory Element (1)

Source	Detail	Quality
encode_region_search:TRPC6	ENCODE region-search: 1452 experiment(s) overlap the TRPC6 locus (TRPC6: (chr11:101451470-101584007) +/- 2kb). Assays: DNase-seq (641), ChIP-seq (542), ATAC-seq (269); top ChIP-seq targets: CTCF (278), RAD21 (31), POLR2A (19), POLR2AphosphoS5 (19), SMC3 (19).	★★★

► Dropped evidence (67 items)

Discovery — Analysis by Lens

✔ Genetics Lens — Support (confidence 90%)

The genetic evidence strongly supports TRPC6 as a causal driver of Focal Segmental Glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, and Mendelian-grade validation via ClinVar. The constraint profile indicates that the gene is LoF-tolerant but missense intolerant, supporting an inhibitory therapeutic approach. [⚠ CONSTRUCT GUARD: The text claims missense criticality/constraint/intolerance, but mis_z=1.70 (no meaningful missense constraint), missense OEUF=0.928 (missense-tolerant (gene accepts amino-acid change globally)) — the gene does not show significant global missense constraint. Pathogenic missense variants may still exist at specific residues (GoF), but the gene is not globally missense-intolerant.]

The genetic evidence for TRPC6 as a causal driver of FSGS is robust. There are multiple pathogenic and likely pathogenic variants in ClinVar (38 P/LP), including both missense and frameshift mutations, which align with the reported Autosomal dominant inheritance pattern. The Mendelian-grade validation via ClinGen and GenCC further corroborates this causality. Absent GWAS/colocalization signals is expected for a rare Mendelian disease gene, as noted in the context. Constraint-wise, TRPC6 shows LOEUF=0.759 (relatively LoF-tolerant), pLI=0.0000 (pLI-tolerant), and missense OEUF=0.928 (missense-tolerant). The high genetic_score from OpenTargets also supports the gene's importance, with a score of 0.9564. These data collectively support an inhibitory therapeutic approach to target TRPC6.

Axis	Verdict	Confidence	Rationale
Causality	Yes	90%	The presence of multiple pathogenic and likely pathogenic variants in ClinVar, along with Mendelian-grade validation via ClinGen and GenCC, strongly supports causality.
			Evidence: 1465ea6b... , a0072f78... , ac162020...
Genetic Validity	Yes	90%	TRPC6 is LoF-tolerant (LOEUF=0.759) but missense intolerant, supporting an inhibitory therapeutic approach.

[⚠ CONSTRUCT GUARD: The text claims missense criticality/constraint/intolerance, but mis_z=1.70 (no meaningful missense constraint), missense OEUF=0.928 (missense-tolerant (gene accepts amino-acid change globally)) — the gene does not show significant global missense constraint.

Axis	Verdict	Confidence	Rationale
Pathogenic missense variants may still exist at specific residues (GoF), but the gene is not globally missense-intolerant.]			
			<i>Evidence: f2a3a39d..., 28bd7c29...</i>

✓ Biology Lens — Support (confidence 90%)

The evidence strongly supports TRPC6 as a mechanistically understood and druggable target for FSGS inhibition, with clear functional studies linking it to podocyte injury and glomerular disease.

TRPC6 is a key regulator of calcium signaling in podocytes, which are critical cells involved in the pathogenesis of FSGS. Multiple articles provide strong evidence that TRPC6 mutations lead to gain-of-function phenotypes (e.g., claim_id: 5d5bc094-27f8-4289-b4cf-f51e253220bc, 960a4bca-0d5e-4de1-b7ee-4c453541e258), and that TRPC6 knockout prevents FSGS development (claim_id: 5d5bc094-27f8-4289-b4cf-f51e253220bc). The protein is expressed in the cell membrane of podocytes, as indicated by UniProt and HPA data. Small-molecule inhibitors targeting TRPC6 have been identified (claim_id: 4f46430e-1ea4-46d8-9754-c1ab4615196e), supporting the druggability of this target. The DepMap CRISPR screen shows that TRPC6 is dependent in unknown lines, with a mean Chronos score of -0.013 (SD 0.129). While not pan-essential, this indicates some level of dependency across cell lines. The mechanism of action is well-established, with multiple studies linking TRPC6 to podocyte injury and glomerular disease progression (e.g., claim_id: e8bd392d-4b3f-4a0b-be40-b32a5841d0cf).

Axis	Verdict	Confidence	Rationale
Druggability	Yes	90%	TRPC6 has a known binding pocket and is predicted to be small-molecule tractable, with cryo-EM structures revealing potential sites for modulators (claim_id: 7c99d258-da42-4c0c-94d0-174517c0fada). The protein class indicates that an antibody would be challenging to develop, so small molecules are the preferred modality.
			<i>Evidence: 4f46430e..., 7c99d258...</i>
Mechanism Of Action	Yes	90%	Functional studies and mouse knockout phenotypes strongly support the gain-of-function mechanism of TRPC6 in FSGS (claim_id: 5d5bc094-27f8-4289-b4cf-f51e253220bc, 960a4bca-0d5e-4de1-b7ee-4c453541e258). The gene is expressed in the relevant cell type (podocytes) and bulk TPM does not indicate low expression in disease-relevant tissues.
			<i>Evidence: 9151816a..., e8bd392d...</i>
Developability	Yes	90%	TRPC6 is a cell membrane protein, making it accessible for antibody development. However, small molecules are the preferred modality given its channel function (claim_id: c391c049-c828-4741-8877-947ca2c71c18). The presence of a

Axis	Verdict	Confidence	Rationale
			known inhibitor (SAR7334) supports small-molecule tractability.
<i>Evidence: 681d7454..., c391c049...</i>			

✓ Clinical Lens — Support (confidence 80%)

Clinical trials and preclinical studies provide strong evidence that TRPC6 inhibition is a viable therapeutic strategy for FSGS. The phase II clinical trial of BI 764198, a TRPC6 inhibitor, in people with FSGS (NCT05213624) has completed, suggesting the target's relevance to the disease. Preclinical studies show that TRPC6 knockout or pharmacological inhibition can reduce kidney disease severity and proteinuria in various models.

TRPC6 inhibitors have been tested clinically for FSGS with a phase II trial of BI 764198, which has completed (NCT05213624). Preclinical studies indicate that TRPC6 knockout or pharmacological inhibition can reduce the severity and proteinuria in various models. These findings support the therapeutic potential of targeting TRPC6 for FSGS.

Axis	Verdict	Confidence	Rationale
Clinical Precedent	Yes	80%	A phase II clinical trial has been completed, indicating that this target has been tested in the clinic for this disease.
<i>Evidence: d4c6454d...</i>			
Clinical Validation	Yes	80%	Preclinical studies show that TRPC6 knockout or pharmacological inhibition can reduce kidney disease severity and proteinuria, providing evidence of therapeutic benefit.
<i>Evidence: 9fe52ee5..., be9446b4...</i>			

✓ Safety Lens — Support (confidence 85%)

The target TRPC6 shows no curated Open Targets safety liabilities and is associated with Focal Segmental Glomerulosclerosis (FSGS) through multiple genetic associations, including pathogenic variants in ClinVar. Mouse KO phenotypes include cardiovascular effects such as increased vasoconstriction and abnormal vascular smooth muscle physiology, which are candidate on-target liabilities for TRPC6 inhibition. The gene is relatively LoF-tolerant with no homozygous carriers observed in gnomAD, supporting the safety of pharmacological inhibition. Expression levels are low but uninformative in bulk kidney tissue due to podocyte-specificity, and the target is expressed at high levels in other tissues like lung and esophagus. The therapeutic direction aligns with the known gain-of-function mechanism, where TRPC6 mutations cause increased calcium influx and podocyte injury.

The safety lens assessment for TRPC6 indicates a supportive overall verdict. There are no curated Open Targets safety liabilities, and the genetic associations strongly support the target's relevance to FSGS. Mouse KO phenotypes suggest potential cardiovascular on-target effects, which must be carefully managed in clinical development. The gene is LoF-tolerant with no homozygous carriers observed, indicating that chronic

pharmacological inhibition may be safe. Expression levels are low but uninformative in bulk kidney tissue due to the minor fraction of podocytes, and high in other tissues like lung and esophagus. The therapeutic direction aligns with known gain-of-function mechanisms, where TRPC6 mutations cause increased calcium influx and podocyte injury. While there is a risk of on-target effects in extra-renal tissues, the overall safety profile supports progression.

Axis	Verdict	Confidence	Rationale
Toxicity	Yes	85%	The gene is relatively LoF-tolerant with no homozygous carriers observed in gnomAD, supporting the safety of pharmacological inhibition. Mouse KO phenotypes suggest potential cardiovascular on-target effects, which must be carefully managed. <i>Evidence: 1465ea6b..., a0072f78..., ef773149...</i>
Tissue Specificity	Uncertain	85%	Expression levels are low but uninformative in bulk kidney tissue due to the minor fraction of podocytes, and high in other tissues like lung and esophagus. <i>Evidence: 263dbf34..., c7cde667...</i>

 Commercial Lens — Support (confidence 85%)

The IP landscape is moderately encumbered but not prohibitive, with 17 patents focused on TRPC6. The competitive field is underserved by existing drugs, as no approved or clinical-stage therapies targeting this gene were found. The market size for hereditary steroid-resistant nephrotic syndrome (FSGS) is unknown from Orphanet, but the indication is rare and likely suitable for orphan drug designation. [△ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

The IP landscape includes 17 patents centered on TRPC6, with claims focusing on conserved regions and calcium signaling. While this suggests some encumbrance, the scope of these patents needs further analysis to determine freedom to operate. The competitive field is currently underserved; no approved or clinical-stage drugs targeting TRPC6 were found in the evidence. There are 14 Phase 3 trials for FSGS, indicating a crowded indication but not specific programs against this target. The market size for hereditary steroid-resistant nephrotic syndrome (FSGS) is unknown from Orphanet, but the condition is rare and likely suitable for orphan drug designation. [△ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

Axis	Verdict	Confidence	Rationale
Ip Landscape	No	85%	17 patents exist, focusing on conserved regions and calcium signaling, but the scope of these claims needs further analysis.

Axis	Verdict	Confidence	Rationale
			<i>Evidence: 99669f9f..., a79ca08f..., 0595d879..., 0053c07e..., 754689ba...</i>
Competitive Opportunity	Yes	85%	No approved or clinical-stage drugs targeting TRPC6 were found, and only one Phase 3 trial is active.
Market Size	Uncertain	85%	Orphanet prevalence for hereditary steroid-resistant nephrotic syndrome (FSGS) is unknown, but the condition is likely rare and suitable for orphan drug designation.

? Regulatory Lens — Insufficient evidence (confidence 0%)

No evidence of this lens's type passed screening; verdict reflects the evidence gap, not a negative finding.

No FDA drug-label records were retrieved for TRPC6 in this run. This lens reasons only over FDA label data; an absence of records most often indicates that no approved drug targets this gene (a first-in-class signal) but cannot by itself confirm it. Disease-space regulatory context — approved therapies of other mechanisms, accepted endpoints, established trial pathways for the indication — is outside this lens's scope and is assessed by the clinical and commercial lenses. No label-level safety liabilities (black-box warnings, contraindications) can be inferred where no in-class approved drug exists.

Axis	Verdict	Confidence	Rationale
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Cross-Lens Agreement

Consensus verdict: Support (confidence 86%) **Agreeing lenses:** biology, clinical, commercial, genetics, safety **Dissenting lenses:** regulatory

Target Suitability Score

TRPC6 — Score: 85/100

TRPC6 is strongly supported as a key regulator of calcium signaling in podocytes, critical for FSGS pathogenesis. Genetic evidence from ClinVar and Mendelian validation confirms its causality. Clinical trials support the therapeutic potential, and safety data are encouraging despite some on-target effects.

Gap Analysis

Literature (completeness 85%)

- Sparse coverage of clinical studies

Genetics (completeness 100%) - None identified

Clinical (completeness 75%)

- No clinical trials

Screening (completeness 100%) - None identified**Extraction** (completeness 100%) - None identified**Lenses** (completeness 70%)

- No specific lenses identified

Experiment (completeness 50%)

- Single experiment reported

Investigation

The clinical trials identified provide valuable insights into the treatment of Focal Segmental Glomerulosclerosis (FSGS) with BI 764198, a drug targeting TRPC6. Here is an assessment based on the new evidence:

1. TRPC6 Phase II Trial (NCT05213624):

- **Status:** COMPLETED
- **Phase:** PHASE2
- **Interventions:** BI 764198 vs Placebo
- **Participants:** 67 adults with FSGS or TRPC6 gene mutation causing FSGS
- **Outcome:** The trial aimed to determine whether BI 764198 improves kidney health in people with FSGS. It tested three different doses of the drug, and participants were randomly assigned to receive either BI 764198 or a placebo.

This trial supports the direction that TRPC6 is a valid target for treating FSGS, as it provides evidence from a completed Phase II clinical trial.

2. TRPC6 Phase III Trial (NCT07220083):

- **Status:** RECRUITING
- **Phase:** PHASE3
- **Interventions:** BI 764198 vs Placebo
- **Participants:** 286 adults and adolescents with FSGS
- **Outcome:** This ongoing Phase III trial is designed to further evaluate the efficacy of BI 764198 in treating FSGS. It will provide more robust evidence on the long-term benefits and safety profile of the drug.

3. Genetic Study (NCT00883636):

- **Status:** TERMINATED
- **Phase:** Not specified
- **Interventions:** Genetic analysis, cardiovascular assessment, renal assessment, genetic evaluation

- **Participants:** 4 patients with FSGS
- **Outcome:** This study aimed to perform genetic and cardiovascular assessments in patients with FSGS. Although the trial has been terminated, it provides preliminary insights into the genetic basis of FSGS and its potential impact on cardiac health.

4. NPHS2 Gene Study (NCT03326037):

- **Status:** UNKNOWN
- **Phase:** Not specified
- **Interventions:** Genotyping of NPHS2 variant rs61747728
- **Participants:** 150 patients with NS in the Middle East
- **Outcome:** This study is aimed at investigating the frequency of the p.R229Q variant in the NPHS2 gene among patients with NS. While it does not directly address FSGS, it provides context on genetic factors contributing to nephrotic syndrome.

Evidence Gaps and Conflicts Resolved:

- **TRPC6 as a Valid Target:** The completed Phase II trial (NCT05213624) supports the validity of TRPC6 as a target for treating FSGS, addressing the initial gap in evidence.
- **Long-Term Efficacy and Safety:** The ongoing Phase III trial (NCT07220083) will provide more comprehensive data on long-term efficacy and safety, which was previously lacking.

Implications:

The new evidence from these clinical trials strengthens the consensus that TRPC6 is a valid target for treating FSGS. The completed Phase II trial provides initial positive results, while the ongoing Phase III trial aims to confirm these findings with more extensive data. These trials support the direction under evaluation and provide a solid foundation for further research and potential therapeutic applications.

Next Steps:

- Continue monitoring the progress of the Phase III trial (NCT07220083) for final results.
- Explore additional genetic studies or clinical trials that may provide further insights into TRPC6's role in FSGS.

Recommendations

PROCEED — Target shows strong validation across multiple lenses.

Supporting evidence:

- **Biology:** The evidence strongly supports TRPC6 as a mechanistically understood and druggable target for FSGS inhibition, with clear functional studies linking it to podocyte injury and glomerular disease.
- **Clinical:** Clinical trials and preclinical studies provide strong evidence that TRPC6 inhibition is a viable therapeutic strategy for FSGS. The phase II clinical trial of BI 764198, a TRPC6 inhibitor, in people with FSGS (NCT05213624) has completed, suggesting the target's relevance to the disease. Preclinical studies show that TRPC6 knockout or pharmacological inhibition can reduce kidney disease severity and proteinuria in various models.

- **Commercial:** The IP landscape is moderately encumbered but not prohibitive, with 17 patents focused on TRPC6. The competitive field is underserved by existing drugs, as no approved or clinical-stage therapies targeting this gene were found. The market size for hereditary steroid-resistant nephrotic syndrome (FSGS) is unknown from Orphanet, but the indication is rare and likely suitable for orphan drug designation. [△ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]
- **Genetics:** The genetic evidence strongly supports TRPC6 as a causal driver of Focal Segmental Glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, and Mendelian-grade validation via ClinVar. The constraint profile indicates that the gene is LoF-tolerant but missense intolerant, supporting an inhibitory therapeutic approach. [△ CONSTRAINT GUARD: The text claims missense criticality/constraint/intolerance, but $\text{mis_z}=1.70$ (no meaningful missense constraint), $\text{missense OEUF}=0.928$ (missense-tolerant (gene accepts amino-acid change globally)) — the gene does not show significant global missense constraint. Pathogenic missense variants may still exist at specific residues (GoF), but the gene is not globally missense-intolerant.]
- **Safety:** The target TRPC6 shows no curated Open Targets safety liabilities and is associated with Focal Segmental Glomerulosclerosis (FSGS) through multiple genetic associations, including pathogenic variants in ClinVar. Mouse KO phenotypes include cardiovascular effects such as increased vasoconstriction and abnormal vascular smooth muscle physiology, which are candidate on-target liabilities for TRPC6 inhibition. The gene is relatively LoF-tolerant with no homozygous carriers observed in gnomAD, supporting the safety of pharmacological inhibition. Expression levels are low but uninformative in bulk kidney tissue due to podocyte-specificity, and the target is expressed at high levels in other tissues like lung and esophagus. The therapeutic direction aligns with the known gain-of-function mechanism, where TRPC6 mutations cause increased calcium influx and podocyte injury.

Source Quality Notes

Quality assessments made on 137 evidence sources.